

Verification II_Rate_and_State_Griffith

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1 Objective

The purpose of this verification exercise is to demonstrate that the numerical simulator Verification II_Rate_and_State_Griffith correctly reproduces the results of Garagash (2021). This is done by digitizing the data of Figures 5 and S5 in Garagash (2021) and comparing them to the simulation results produced by the MATLAB and Python simulators.

2 Results

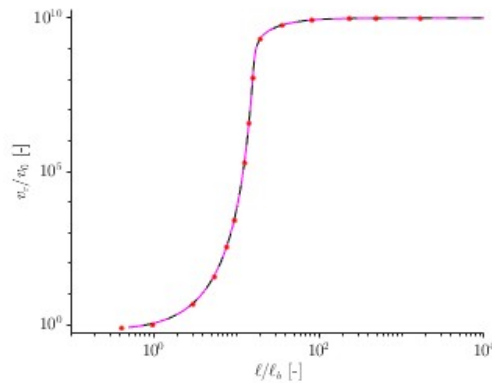


Figure 1: Rupture velocity as function of crack length for MATLAB (black) and Python (dashed pink). The digitized results of Garagash (2021) are shown with red dots. This figure corresponds to the aging-law simulations of Figure S5 of Garagash (2021) in the case of $\Delta T = 0$, $a/b = 0.9$.

3 Conclusion

The simulators have both been verified.

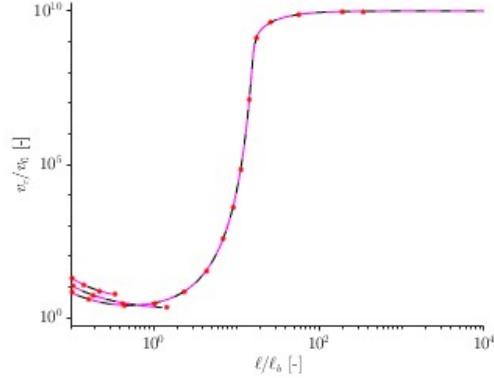


Figure 2: Rupture velocity as function of crack length for MATLAB (black) and Python (dashed pink). Each line corresponds to a different background stress. The digitized results of Garagash (2021) are shown with red dots. This figure corresponds to the aging-law simulations of Figure S5 of Garagash (2021) in the case of $\Delta T = 2$, $a/b = 0.9$.

References

- D. Garagash. Fracture mechanics of rate-and-state faults and fluid injection induced slip. *Philosophical Transactions of the Royal Society A*, 379, 2021. doi: <https://doi.org/10.1098/rsta.2020.0129>.

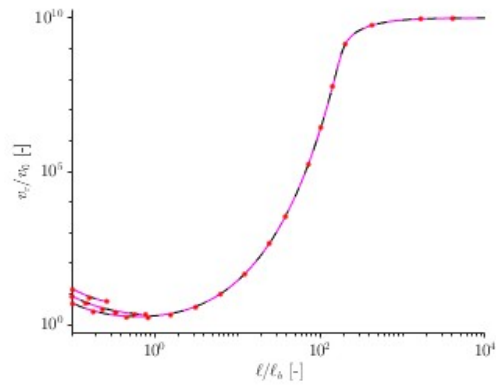


Figure 3: Rupture velocity as function of crack length for MATLAB (black) and Python (dashed pink). Each line corresponds to a different background stress. The digitized results of Garagash (2021) are shown with red dots. This figure corresponds to the aging-law simulations of Figure S5 of Garagash (2021) in the case of $\Delta T = 2$, $a/b = 1.0$.